



More k-space data (higher matrix, larger bandwidth) → better spatial resolution and sharper images.



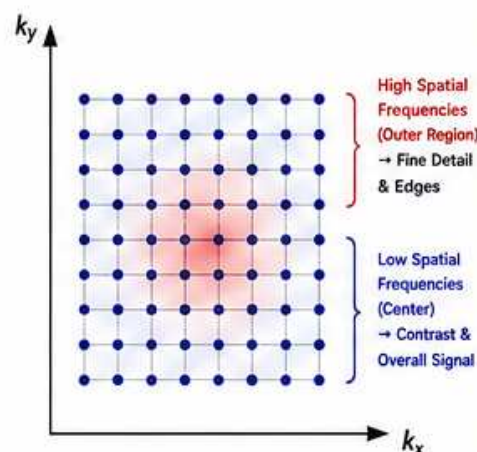
THE BIG IDEA

Spatial resolution (sharpness and detail) depends primarily on how much of k-space is sampled and how finely it is sampled.

THE CONCEPT

- 1 K-space extends from the center (low spatial frequencies) to the edges (high spatial frequencies).
- 2 High spatial frequencies (outer k-space) contain the fine details and sharp edges of the image.
- 3 Sampling more of k-space (especially the outer region) improves spatial resolution.
- 4 Higher matrix = more samples in k-space → finer detail.
- 5 Higher receiver bandwidth = wider k-space → sharper edges in the frequency (k_x) direction.

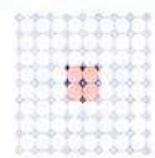
K-SPACE MAP (2D)



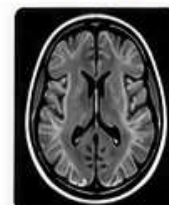
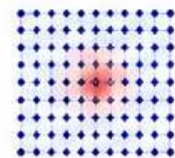
RESOLUTION IS DETERMINED BY K-SPACE COVERAGE & SAMPLING DENSITY

A. K-SPACE COVERAGE (FOV)

Larger FOV samples a smaller range of k-space → lower resolution.
Smaller FOV samples a larger range of k-space → higher resolution.

LARGE FOV
(SMALL K-SPACE COVERAGE)

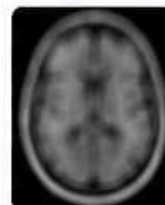
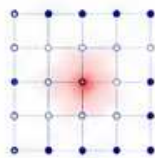
Lower Resolution
Less detail

SMALL FOV
(LARGE K-SPACE COVERAGE)

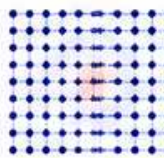
Higher Resolution
More detail

B. K-SPACE SAMPLING DENSITY (MATRIX)

Higher matrix = more samples within the same k-space range → better spatial resolution.

LOW MATRIX
(128 × 128)

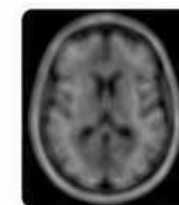
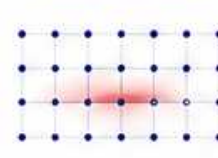
Lower Resolution
Blocky, blurry

HIGH MATRIX
(512 × 512)

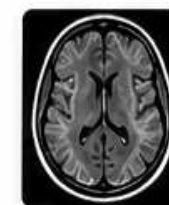
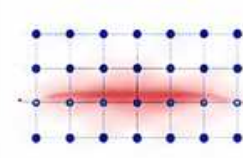
Higher Resolution
Sharper, more detail

C. RECEIVER BANDWIDTH (RBW)

Higher bandwidth samples a wider range of k_x → sharper edges in the left-right direction.

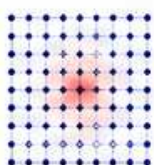
LOW BANDWIDTH
(NARROW k_x RANGE)

Softer edges
More blurring (readout direction)

HIGH BANDWIDTH
(WIDE k_x RANGE)

Sharper edges
Better definition (readout direction)

K-SPACE & IMAGE RELATIONSHIP

K-SPACE
(What we sample)IMAGE
(What we see)

● Center (Low Frequencies)

- Controls contrast and overall signal
- Loss → low contrast

● Outer (High Frequencies)

- Controls edges and fine detail
- Loss → blurry, soft, no detail

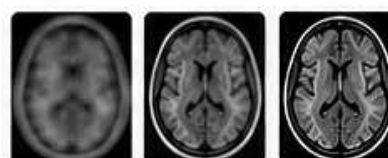
● Full K-Space

- Both contrast and detail are preserved
- Best overall image quality

EXAMPLES: HOW RESOLUTION CHANGES

Matrix (Same FOV)

128 × 128 256 × 256 512 × 512



Increasing Matrix → Higher Resolution →
(More k-space samples)

FOV (Same Matrix)

Large FOV (24 cm) Medium FOV (18 cm) Small FOV (12 cm)



Decreasing FOV → Higher Resolution →
(More k-space coverage)

Bandwidth (Same Matrix & FOV)

Low RBW (±15 kHz) Medium RBW (±31 kHz) High RBW (±62 kHz)



Increasing Bandwidth → Sharper Edges →
(Wider k_x sampling)

THE TRADE-OFFS



Higher Matrix

- + Better resolution (more detail)
- Longer scan time, lower SNR

Smaller FOV

- + Better resolution
- Risk of aliasing (wrap-around)

Higher Bandwidth

- + Sharper edges
- Lower SNR

More K-Space Data

- + Better image quality
- Longer scan time

KEY FORMULA (1D APPROXIMATION)

Pixel Size (Δx)

$$\Delta x = \text{FOV}_x / N_x$$

(Smaller pixel size
→ higher resolution)

K-Space Sampling Interval (Δk_x)

$$\Delta k_x = \frac{1}{\text{FOV}_x}$$

(Smaller Δk_x → larger FOV)

K-Space Extent ($K_{x,\text{max}}$)

$$K_{x,\text{max}} = N_x \cdot \Delta k_x = \frac{N_x}{2 \cdot \text{FOV}_x}$$

(Larger extent → smaller pixel size
→ higher resolution)

KEY TAKEAWAYS

- ✓ Image sharpness comes from sampling high spatial frequencies (outer k-space).
- ✓ Higher matrix, smaller FOV, and higher bandwidth all improve resolution.
- ✓ More k-space data = better detail, but at the cost of time or SNR.
- ✓ The best images balance resolution, SNR, scan time, and coverage.

